

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

ME 19-205-0406 MANUFACTURING PROCESSES
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Learn about the sand casting process.

CO2: Understand the special casting processes and casting defects.

CO3: Learn about different metal forming processes.

CO4: Understand different metal joining processes and their characteristics.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) What are the specific advantages of the top gate and what are its applications?		3	L2	1	1,2
(b) Briefly explain aspiration effect in sand moulding.		3	L1	1	1
(c) Specify the advantages of the precision investment casting process over other casting processes.		3	L2	2	1,2
(d) How will you compare a cold chamber die casting process with a hot chamber process?		3	L2	2	1,2
(e) State the Tresca's maximum shear stress criterion for yielding in metal forming process.		3	L1	3	1,2
(f) What is indirect extrusion?		3	L1	3	1,2
(g) What is meant by penetration in welding?		3	L2	4	1,2
(h) How is an arc obtained in arc welding?		3	L2	4	1,2

PART B

(4 × 12 = 48)

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II. (a)	Name two allowances provided on the pattern for sand casting and state the reasons why they are provided.	4	L1	1	2,3
(b)	Explain the desirable characteristics of core in sand casting.	4	L2	1	2,3
(c)	A rectangular prism casting has dimensions of 150 mm × 100 mm × 50 mm. If the mold constant is 0.2, calculate the solidification time.	4	L3	1	2,3
OR					
III. (a)	Distinguish clearly between moulding sand, backing sand and facing sand.	4	L2	1	2,3
(b)	Describe the allowances given on a pattern for shaking and distortion.	4	L2	1	2,3
(c)	Explain how the solidification of pure metals different from alloys in sand molding.	4	L3	1	2,3

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Briefly explain centrifugal casting with the help of sketches.	6	L2	2	1,2
	(b) Calculate the size of a cylindrical riser (height and diameter equal) necessary to feed a steel slab casting $25 \times 25 \times 5$ cm, with a top riser casting poured horizontally into the mould (Use Caine's method for solution; for steel, constants $a = 0.1$, $b = 0.03$, $c = 1.00$).	6	L3	2	2,3
OR					
V.	(a) Explain how do you calculate the time taken to fill up the mould in vertical gating.	6	L1	2	2
	(b) Briefly explain casting defects. Give suitable sketches.	6	L1	2	1,2
VI.	(a) What are the different sheet metal forming processes? Explain with sketches.	5	L2	3	1,2
	(b) A 300 mm wide strip is fed through a rolling mill with two powered rolls each of radius 250 mm. The work thickness is to be reduced to 22 mm in one pass at a roll speed of 50 rpm. The work material has a flow curve defined by $K = 275$ MPa and $n = 0.15$ and the coefficient of friction between the rolls and the work is 0.12. Determine if the friction is sufficient to permit the rolling operation to be accomplished. If so, calculate the roll force.	7	L4	3	2,3
OR					
VII.	(a) What is spring back in sheet metal bending? Explain.	5	L2	3	1,2
	(b) A cup drawing operation is performed in which the inside dia = 80 mm and height = 50 mm. The stock thickness = 3 mm and the starting blank dia = 150 mm, punch radius = 4 mm, Tensile strength = 400 MPa and yield strength = 180 MPa for the sheet metal. Determine the drawing ratio, reduction and the drawing force.	7	L4	3	2,3
VIII	(a) Describe the oxy-acetylene gas welding technique.	5	L2	4	1,2
	(b) What is flux? Why is it essential to use it in some welding situations?	3	L2	4	1,2
	(c) Explain the necessity of inert gas shielded arc welding processes.	4	L2	4	2
OR					
IX.	(a) Show with the help of a schematic sketch, a resistance welding operation.	6	L2	4	1,2
	(b) Discuss the heat flow and its consequences in the case of an arc welded joint.	6	L2	4	2,3

Bloom's Taxonomy Levels

L1 – 20.77%, L2 – 55.83%, L3 – 11.7%, L4 – 11.7%.
